Dual-IMU AHRS Simulation Report

Quaternion EKF and Mahony Filter Comparison

Simulation study using a single and dual MARG/IMU sensor model

June 19, 2026

Contents

1 Executive Summary

The uploaded result package is internally consistent and the estimator behavior is physically reasonable. No code-level fixes are required before writing up the project. The quaternion norms remain at machine precision, the magnetometer calibration reduces direction error, and the dual-IMU disagreement metrics are nonzero and bounded, which confirms that the dual-sensor path is active rather than accidentally duplicating the single-IMU path.

The best-performing case is the **Dual-IMU EKF**. In the single main run, its roll, pitch, and yaw RMSE values are

$$0.0905^\circ, \qquad 0.0695^\circ, \qquad 0.1701^\circ,$$

respectively. Across 50 Monte Carlo runs, the same case gives mean roll, pitch, and yaw RMSE values of

$$0.0855^\circ, \qquad 0.0708^\circ, \qquad 0.1736^\circ.$$

The dual-IMU implementation improves both EKF and Mahony performance. The EKF remains more accurate than Mahony for the present gain tuning. Mahony is stable and useful as a lightweight AHRS baseline, but the EKF has better attitude accuracy and better gyro-bias behavior in this simulation.

2 Notation and Frame Convention

The navigation frame is denoted by n, and the body frame by b. The attitude quaternion is written as

$$\mathbf{q}_{nb} = \begin{bmatrix} q_w & q_x & q_y & q_z \end{bmatrix}^T,$$

where $\mathbf{q}_{nb}$ maps body-frame vectors into the navigation frame through the rotation matrix $\mathbf{R}_{nb} = \mathbf{R}(\mathbf{q}_{nb})$. Therefore,

$$\mathbf{v}^n = \mathbf{R}_{nb}\mathbf{v}^b, \qquad \mathbf{v}^b = \mathbf{R}_{nb}^T\mathbf{v}^n.$$

Quaternion multiplication is denoted by $\otimes$. For an angular-rate vector $\boldsymbol{\omega}^b$, the pure quaternion form is

$$\boldsymbol{\omega}_q = \begin{bmatrix} 0 & \omega_x & \omega_y & \omega_z \end{bmatrix}^T.$$

The quaternion kinematics used throughout the simulation are

$$\dot{\mathbf{q}}_{nb} = \frac{1}{2}\mathbf{q}_{nb} \otimes \boldsymbol{\omega}_q.$$

After numerical integration, the quaternion is normalized to maintain

$$\|\mathbf{q}_{nb}\| = 1.$$

3 IMU and Magnetometer Modeling

3.1 Ideal specific force and magnetic field

An accelerometer does not directly measure navigation-frame acceleration. It measures *specific force*, which is acceleration relative to gravity. If the true navigation-frame translational acceleration is $\mathbf{a}^n$, and gravity is $\mathbf{g}^n$, then the ideal body-frame accelerometer signal is

$$\mathbf{f}^b = \mathbf{R}_{nb}^T\left(\mathbf{a}^n - \mathbf{g}^n\right). \tag{1}$$

For an attitude-only simulation, the accelerometer is mainly used as a gravity-direction sensor. When the body is not undergoing strong external acceleration, $\mathbf{f}^b$ is dominated by the gravity direction expressed in the body frame.

The ideal magnetometer measurement is modeled as a known local magnetic field vector $\mathbf{m}^n$ rotated into the body frame:

$$\mathbf{m}^b = \mathbf{R}_{nb}^T \mathbf{m}^n. \tag{2}$$

The magnetometer gives heading information because yaw rotation changes the horizontal projection of $\mathbf{m}^b$, whereas accelerometer gravity measurements alone cannot observe yaw.

3.2 Gyroscope model

The simulated gyroscope measurement is represented as

$$\tilde{\boldsymbol{\omega}}_k^b = \mathbf{S}_g \boldsymbol{\omega}_k^b + \mathbf{b}_{g\,k} + \mathbf{n}_{g,k}, \tag{3}$$

where $\mathbf{S}_g$ is the diagonal scale-factor matrix, $\mathbf{b}_{g\,k}$ is the gyro bias, and $\mathbf{n}_{g,k}$ is zero-mean white noise. The gyro bias is updated as a random walk:

$$\mathbf{b}_{g\,k+1} = \mathbf{b}_{g\,k} + \boldsymbol{\sigma}_{bg} \sqrt{\Delta t}\, \boldsymbol{\eta}_{g,k}, \qquad \boldsymbol{\eta}_{g,k} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \tag{4}$$

This is a standard low-cost IMU approximation: the gyro is accurate over short time windows, but any residual bias produces attitude drift if the measurement is simply integrated.

3.3 Accelerometer model

The accelerometer measurement model is

$$\tilde{\mathbf{f}}_k^b = \mathbf{S}_a \mathbf{f}^b{}_k + \mathbf{b}_{a\,k} + \mathbf{n}_{a,k}, \tag{5}$$

with random-walk bias

$$\mathbf{b}_{a\,k+1} = \mathbf{b}_{a\,k} + \boldsymbol{\sigma}_{ba} \sqrt{\Delta t}\, \boldsymbol{\eta}_{a,k}, \qquad \boldsymbol{\eta}_{a,k} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \tag{6}$$

For the AHRS correction step, the accelerometer is normalized:

$$\mathbf{z}_{a,k} = \frac{\tilde{\mathbf{f}}_k^b}{\left\| \tilde{\mathbf{f}}_k^b \right\|}. \tag{7}$$

Only the direction is used in the attitude correction, which makes the update less sensitive to the scale of gravity.

3.4 Magnetometer model and calibration

The raw magnetometer model is

$$\tilde{\mathbf{m}}_k^b = \mathbf{S}_m \mathbf{m}_k^b + \mathbf{b}_{m\,k} + \mathbf{n}_{m,k}. \tag{8}$$

The project applies a known hard-iron and scale-factor calibration before passing the magnetometer to the AHRS:

$$\hat{\mathbf{m}}_k^b = \mathbf{S}_m^{-1} \left(\tilde{\mathbf{m}}_k^b - \mathbf{b}_m \right). \tag{9}$$

The calibrated vector is then normalized:

$$\mathbf{z}_{m,k} = \frac{\hat{\mathbf{m}}_k^b}{\left\| \hat{\mathbf{m}}_k^b \right\|}. \tag{10}$$

The results show that this calibration matters. In the single-IMU cases, the magnetic direction RMSE is reduced from 1.5570° to 0.8077°. In the dual-IMU cases, the raw fused magnetic direction RMSE is 0.7154°, and the calibrated value is 0.6221°.

3.5 Saturation and quantization

The sensor output is also bounded and quantized. For a sensor range r, the saturation operation is

$$\mathrm{sat}(y, r) = \min\left(\max(y, -r), r\right). \tag{11}$$

For N bits, the quantization step is

$$\Delta_q = \frac{2r}{2^N}, \tag{12}$$

so the quantized output is

$$y_q = \Delta_q \,\mathrm{round}\left(\frac{\mathrm{sat}(y, r)}{\Delta_q}\right). \tag{13}$$

4 EKF Attitude Estimation

4.1 State vector

The EKF estimates attitude and gyro bias. The state vector is

$$\mathbf{x}_k = \begin{bmatrix} q_w & q_x & q_y & q_z & b_{gx} & b_{gy} & b_{gz} \end{bmatrix}^T. \tag{14}$$

The quaternion carries the attitude, and the bias state compensates slowly varying gyro drift.

4.2 Prediction step

The corrected angular rate is

$$\hat{\boldsymbol{\omega}}_k^b = \tilde{\boldsymbol{\omega}}_k^b - \hat{\mathbf{b}_g}_k. \tag{15}$$

A first-order quaternion integration gives

$$\mathbf{q}_{nb}{}_{k+1}^{-} = \mathrm{norm}\left(\mathbf{q}_{nb}{}_k^{+} + \frac{1}{2}\left(\mathbf{q}_{nb}{}_k^{+} \otimes \begin{bmatrix} 0 \\ \hat{\boldsymbol{\omega}}_k^b \end{bmatrix}\right)\Delta t\right), \tag{16}$$

while the gyro bias is modeled as constant over one time step:

$$\hat{\mathbf{b}_g}_{k+1}^{-} = \hat{\mathbf{b}_g}_k^{+}. \tag{17}$$

The nonlinear state transition is written compactly as

$$\mathbf{x}_{k+1}^{-} = \mathbf{f}\left(\mathbf{x}_k^{+}, \tilde{\boldsymbol{\omega}}_k^b\right). \tag{18}$$

The covariance is propagated with the linearized state-transition matrix

$$\mathbf{F}_k = \left.\frac{\partial \mathbf{f}}{\partial \mathbf{x}}\right|_{\mathbf{x}_k^{+}, \tilde{\boldsymbol{\omega}}_k^b}. \tag{19}$$

The implementation uses a numerical Jacobian for clarity. The prediction covariance update is

$$\mathbf{P}_{k+1}^{-} = \mathbf{F}_k \mathbf{P}_k^{+} \mathbf{F}_k^T + \mathbf{Q}. \tag{20}$$

4.3 Measurement model

The EKF uses two vector observations: gravity direction from the accelerometer and magnetic-field direction from the magnetometer. The normalized measurement vector is

$$\mathbf{z}_k = \begin{bmatrix} \mathbf{z}_{a,k} \\ \mathbf{z}_{m,k} \end{bmatrix} = \begin{bmatrix} \tilde{\mathbf{f}}_k^b / \left\| \tilde{\mathbf{f}}_k^b \right\| \\ \hat{\mathbf{m}}_k^b / \left\| \hat{\mathbf{m}}_k^b \right\| \end{bmatrix}. \tag{21}$$

The predicted gravity and magnetic vectors in the body frame are

$$\hat{\mathbf{g}}_k^b = \mathbf{R}_{nb}(\hat{\mathbf{q}_{nbk}})^T \mathbf{e}_g, \qquad \hat{\mathbf{m}}_k^b = \mathbf{R}_{nb}(\hat{\mathbf{q}_{nbk}})^T \mathbf{m}^n, \tag{22}$$

where $\mathbf{e}_g$ is the normalized gravity reference direction and $\mathbf{m}^n$ is the normalized navigation-frame magnetic field. The EKF measurement function is therefore

$$\mathbf{h}(\mathbf{x}_k) = \begin{bmatrix} \mathbf{R}_{nb}(\mathbf{q}_{nbk})^T \mathbf{e}_g \\ \mathbf{R}_{nb}(\mathbf{q}_{nbk})^T \mathbf{m}^n \end{bmatrix}. \tag{23}$$

The corresponding measurement Jacobian is

$$\mathbf{H}_k = \left. \frac{\partial \mathbf{h}}{\partial \mathbf{x}} \right|_{\mathbf{x}_k^-}. \tag{24}$$

Again, the implementation uses a numerical Jacobian. This is computationally slower than an analytic Jacobian, but it is readable and adequate for the simulation study.

4.4 Correction step

The innovation is

$$\mathbf{r}_k = \mathbf{z}_k - \mathbf{h}(\mathbf{x}_k^-). \tag{25}$$

The innovation covariance and Kalman gain are

$$\mathbf{S}_k = \mathbf{H}_k \mathbf{P}_k^- \mathbf{H}_k^T + \mathbf{R}, \qquad \mathbf{K}_k = \mathbf{P}_k^- \mathbf{H}_k^T \mathbf{S}_k^{-1}. \tag{26}$$

The state update is

$$\mathbf{x}_k^+ = \mathbf{x}_k^- + \mathbf{K}_k \mathbf{r}_k. \tag{27}$$

The covariance uses the Joseph stabilized form:

$$\mathbf{P}_k^+ = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k)\, \mathbf{P}_k^- \,(\mathbf{I} - \mathbf{K}_k \mathbf{H}_k)^T + \mathbf{K}_k \mathbf{R} \mathbf{K}_k^T. \tag{28}$$

Finally, the quaternion part of the state is renormalized.

4.5 Measurement gating

The update is applied only if the measured accelerometer and magnetometer norms lie inside admissible gates:

$$\alpha_{a,\min} < \left\| \tilde{\mathbf{f}}_k^b \right\| < \alpha_{a,\max}, \qquad \alpha_{m,\min} < \left\| \hat{\mathbf{m}}_k^b \right\| < \alpha_{m,\max}. \tag{29}$$

This prevents gross accelerometer or magnetometer disturbances from entering the correction step.

5 Mahony AHRS Implementation

The Mahony filter is a nonlinear complementary filter on attitude. In this project it is implemented with the same external interface as the EKF: it receives gyroscope, accelerometer, and magnetometer measurements and returns a quaternion and a gyro-bias estimate. The main difference is that Mahony uses deterministic proportional-integral feedback instead of a covariance-based Kalman update.

5.1 Predicted reference directions

At time step k, the filter predicts the body-frame gravity and magnetic directions from the current quaternion:

$$\hat{\mathbf{g}}_k^b = \mathbf{R}_{nb}(\hat{\mathbf{q}_{nbk}})^T \mathbf{e}_g, \qquad \hat{\mathbf{m}}_k^b = \mathbf{R}_{nb}(\hat{\mathbf{q}_{nbk}})^T \mathbf{m}^n. \tag{30}$$

The measured unit vectors are

$$\mathbf{z}_{a,k} = \frac{\tilde{\mathbf{f}}_k^b}{\left\|\tilde{\mathbf{f}}_k^b\right\|}, \qquad \mathbf{z}_{m,k} = \frac{\hat{\mathbf{m}}_k^b}{\left\|\hat{\mathbf{m}}_k^b\right\|}. \tag{31}$$

5.2 Attitude correction error

The attitude error is formed from vector cross-products:

$$\mathbf{e}_k = \mathbf{z}_{a,k} \times \hat{\mathbf{g}}_k^b + \mathbf{z}_{m,k} \times \hat{\mathbf{m}}_k^b. \tag{32}$$

Each cross product is small when the measured and predicted directions agree. The accelerometer term primarily corrects roll and pitch; the magnetometer term contributes yaw observability.

5.3 Bias and gyro correction

The gyro-bias estimate is updated by the integral feedback law

$$\hat{\mathbf{b}}_{g_{k+1}} = \hat{\mathbf{b}}_{g_k} - K_i \mathbf{e}_k \Delta t. \tag{33}$$

The corrected angular rate used for quaternion propagation is

$$\hat{\boldsymbol{\omega}}_k^b = \tilde{\boldsymbol{\omega}}_k^b - \hat{\mathbf{b}}_{g_k} + K_p \mathbf{e}_k. \tag{34}$$

The quaternion is propagated using the same kinematic equation as the EKF:

$$\hat{\mathbf{q}_{nbk+1}} = \operatorname{norm}\left(\hat{\mathbf{q}_{nbk}} + \frac{1}{2}\left(\hat{\mathbf{q}_{nbk}} \otimes \begin{bmatrix} 0 \\ \hat{\boldsymbol{\omega}}_k^b \end{bmatrix}\right) \Delta t\right). \tag{35}$$

The gains K_p and K_i control the transient correction and bias convergence. Larger K_p gives a stronger attitude correction but can make the response noisier. Larger K_i makes bias convergence faster but can also amplify low-frequency measurement disturbances.

6 Dual-IMU Sensor-Fusion Logic

6.1 Assumptions

The current dual-IMU logic assumes both units are aligned with the same body frame and mounted without a significant lever arm. Under that assumption, both IMUs observe the same ideal angular rate, specific force, and magnetic vector. Their measurement errors differ because each unit has independent noise, bias, scale factor, and magnetometer calibration parameters.

If the sensors were physically separated by non-negligible vectors $\mathbf{r}_1$ and $\mathbf{r}_2$, the accelerometer model would need lever-arm compensation:

$$\mathbf{f}_i^b = \mathbf{f}_0^b + \dot{\boldsymbol{\omega}}^b \times \mathbf{r}_i + \boldsymbol{\omega}^b \times \left(\boldsymbol{\omega}^b \times \mathbf{r}_i\right). \tag{36}$$

That effect is not included in the present project, which is appropriate for the co-located/aligned dual-IMU simulation.

6.2 Weighted measurement fusion

The project fuses the two sensor outputs before passing them to the AHRS. For a generic three-axis measurement $\mathbf{y}$, with two independent measurements $\mathbf{y}_1$ and $\mathbf{y}_2$, each axis is fused with inverse-variance weighting:

$$\mathbf{w}_1 = \frac{1}{\boldsymbol{\sigma}_1^2}, \qquad \mathbf{w}_2 = \frac{1}{\boldsymbol{\sigma}_2^2}, \tag{37}$$

where the division is elementwise. The fused measurement is

$$\mathbf{y}_f = \frac{\mathbf{w}_1 \odot \mathbf{y}_1 + \mathbf{w}_2 \odot \mathbf{y}_2}{\mathbf{w}_1 + \mathbf{w}_2}. \tag{38}$$

This is applied to gyroscope, accelerometer, and calibrated magnetometer measurements. The expected benefit is that independent white noise is reduced. If two sensors have equal variance σ^2, then simple averaging gives

$$\sigma_f = \frac{\sigma}{\sqrt{2}}. \tag{39}$$

With unequal noise levels, the inverse-variance formula naturally gives more weight to the cleaner sensor.

6.3 Disagreement metrics

The dual-IMU health metrics are the instantaneous disagreement norms:

$$d_{a,k} = \left\|\tilde{\mathbf{f}}_{1,k} - \tilde{\mathbf{f}}_{2,k}\right\|, \quad d_{g,k} = \|\tilde{\boldsymbol{\omega}}_{1,k} - \tilde{\boldsymbol{\omega}}_{2,k}\|, \quad d_{m,k} = \left\|\hat{\mathbf{m}}^b_{1,k} - \hat{\mathbf{m}}^b_{2,k}\right\|. \tag{40}$$

The reported RMS values are

$$d_{\text{rms}} = \sqrt{\frac{1}{N}\sum_{k=1}^{N} d_k^2}. \tag{41}$$

The dual cases show nonzero disagreement values, confirming that the dual path is using two distinct simulated sensors.

7 Results

7.1 Main-run comparison

Table 1 summarizes the four main simulation cases. The dual EKF gives the best attitude RMSE in the single main run. Mahony is stable, but the EKF performs better with the present tuning.

Table 1: Main-run attitude results for all four cases.

Case	Roll RMSE [deg]	Pitch RMSE [deg]	Yaw RMSE [deg]	Bias error norm [deg/s]
Single IMU EKF	0.2286	0.1613	0.4370	0.1112
Dual IMU EKF	**0.0905**	**0.0695**	**0.1701**	0.2097
Single IMU Mahony	0.2899	0.2320	0.7022	0.3828
Dual IMU Mahony	0.1817	0.1197	0.4934	0.2874

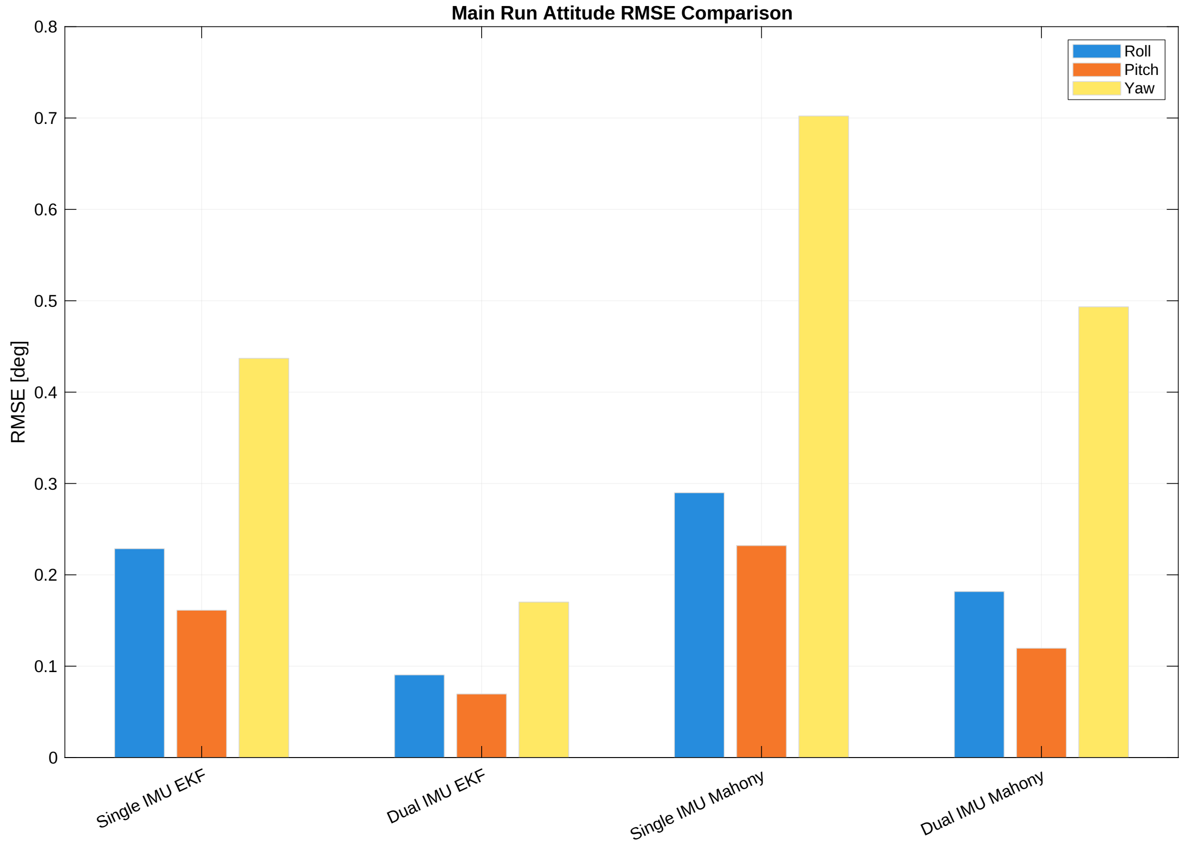


Figure 1: Main-run attitude RMSE comparison across the four cases.

Relative to Single IMU EKF, Dual IMU EKF reduces main-run RMSE by approximately 60.4% in roll, 56.9% in pitch, and 61.1% in yaw. Relative to Single IMU Mahony, Dual IMU Mahony reduces main-run RMSE by approximately 37.3%, 48.4%, and 29.7%, respectively.

7.2 Monte Carlo comparison

The Monte Carlo study uses 50 runs per case. Table 2 reports the mean and standard deviation of attitude RMSE.

Table 2: Monte Carlo attitude RMSE over 50 runs. Values are mean ± standard deviation.

Case	Roll RMSE [deg]	Pitch RMSE [deg]	Yaw RMSE [deg]
Single IMU EKF	0.1968 ± 0.0125	0.1584 ± 0.0137	0.4333 ± 0.0343
Dual IMU EKF	$\mathbf{0.0855 \pm 0.0075}$	$\mathbf{0.0708 \pm 0.0038}$	$\mathbf{0.1736 \pm 0.0118}$
Single IMU Mahony	0.2944 ± 0.0590	0.2952 ± 0.0770	1.1752 ± 0.3679
Dual IMU Mahony	0.2045 ± 0.0433	0.1773 ± 0.0503	0.6905 ± 0.3342

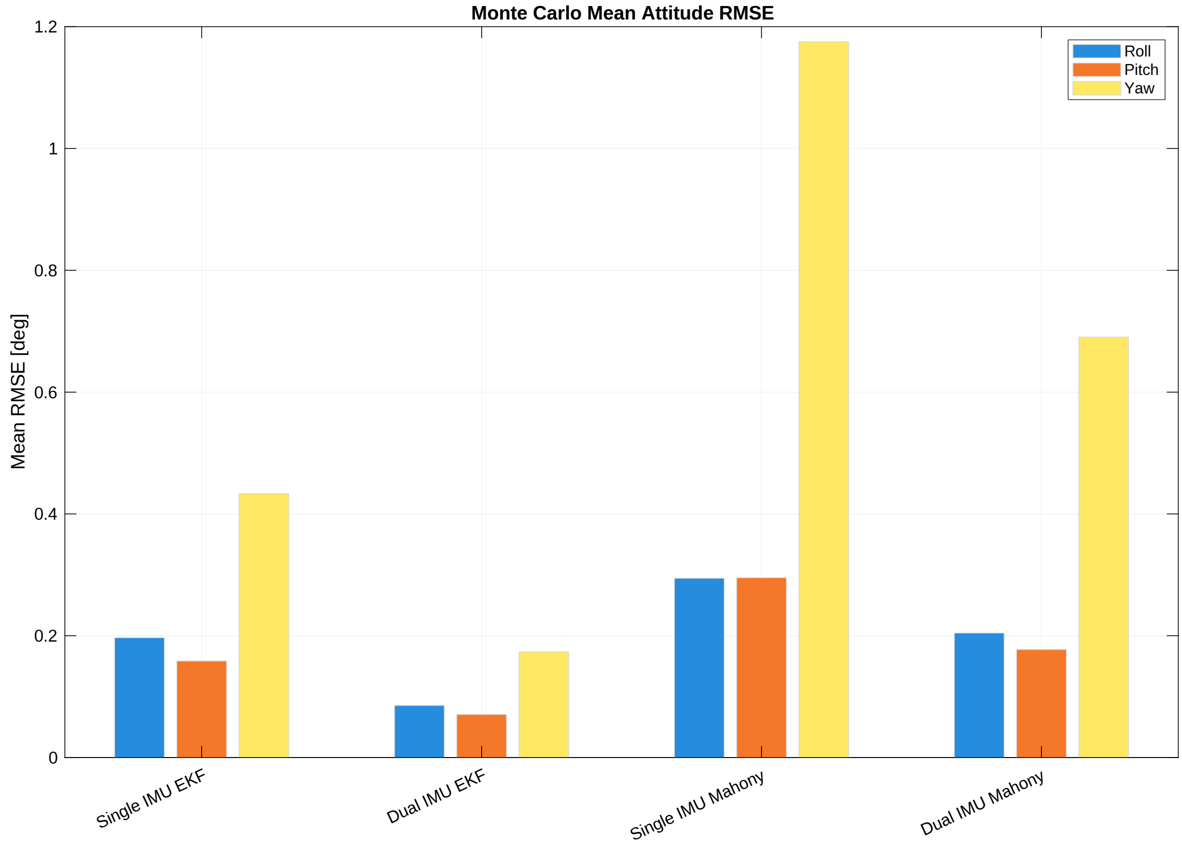


Figure 2: Monte Carlo mean attitude RMSE for all four cases.

The Monte Carlo data confirms the main-run behavior. In the EKF case, moving from single IMU to dual IMU reduces the mean RMSE by approximately 56.5% in roll, 55.3% in pitch, and 59.9% in yaw. In the Mahony case, the same change reduces the mean RMSE by approximately 30.5%, 39.9%, and 41.2%, respectively.

The EKF is also more consistent than Mahony under the current filter gains, as shown in Fig. 3. The Mahony yaw RMSE has a noticeably larger spread across Monte Carlo runs.

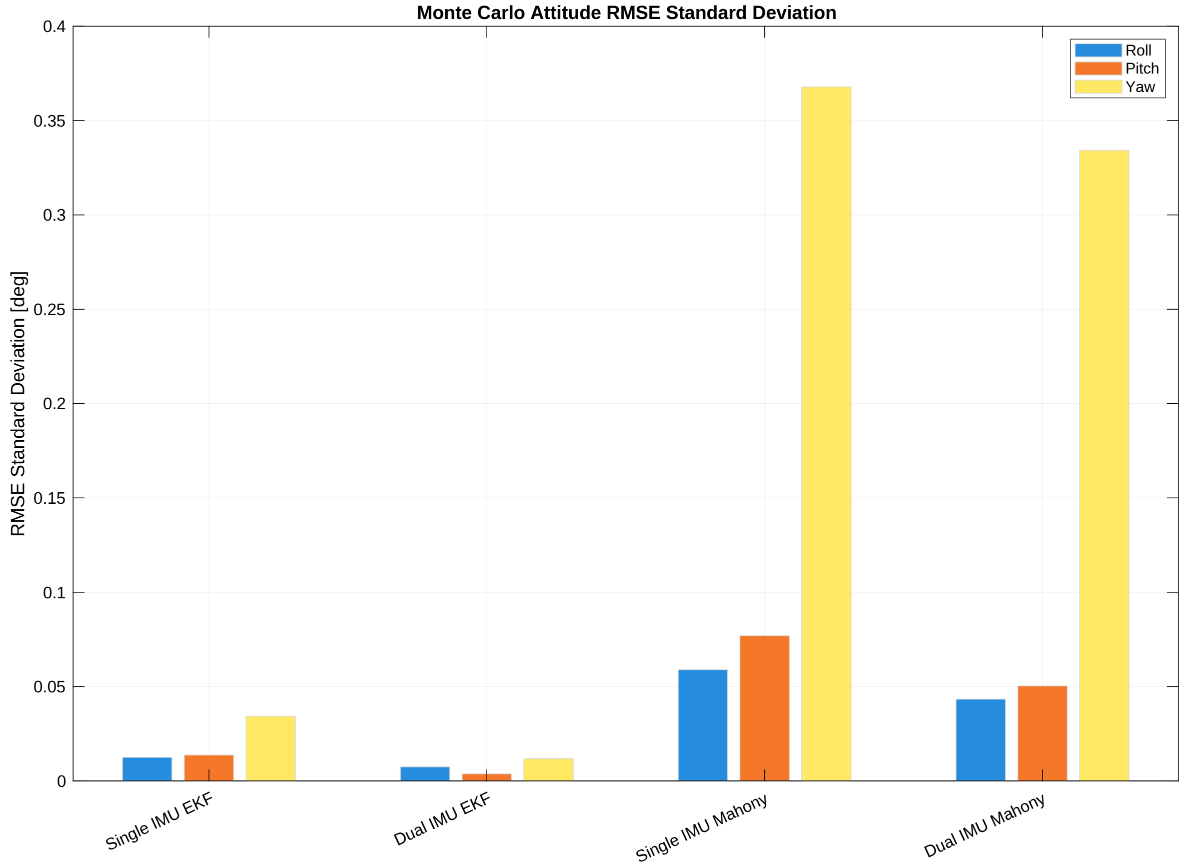


Figure 3: Monte Carlo standard deviation of the attitude RMSE.

7.3 Filter comparison

For the dual-IMU configuration, EKF is significantly more accurate than Mahony. In the Monte Carlo mean values, Dual EKF has lower RMSE than Dual Mahony by approximately 58.2% in roll, 60.1% in pitch, and 74.9% in yaw. This is consistent with the EKF using a measurement covariance model and explicit bias state covariance propagation.

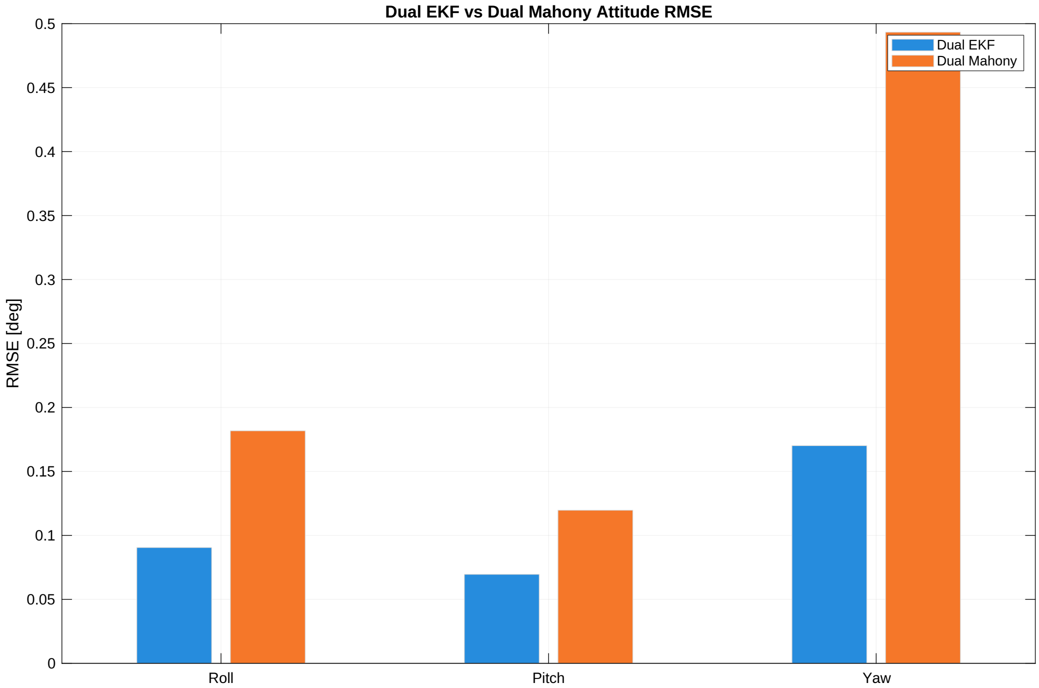
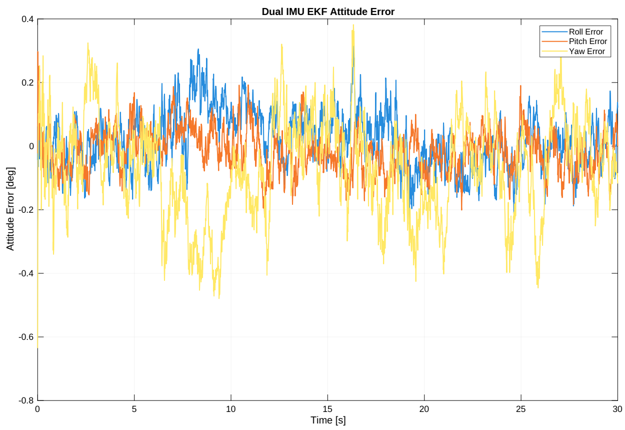


Figure 4: Dual-IMU EKF versus Dual-IMU Mahony attitude RMSE in the main run.

Figure 5 compares the time-domain attitude errors for the dual EKF and dual Mahony main runs. Both filters remain bounded, but the EKF is tighter.



(a) Dual EKF.

(b) Dual Mahony.

Figure 5: Time-domain attitude error for the two dual-IMU filters.

7.4 Gyro-bias and magnetometer behavior

The Monte Carlo mean final gyro-bias error norm is lowest for the dual EKF:

$$0.1896 \text{ deg/s}.$$

The Mahony bias error is larger under the current gains, especially in the single-IMU case.

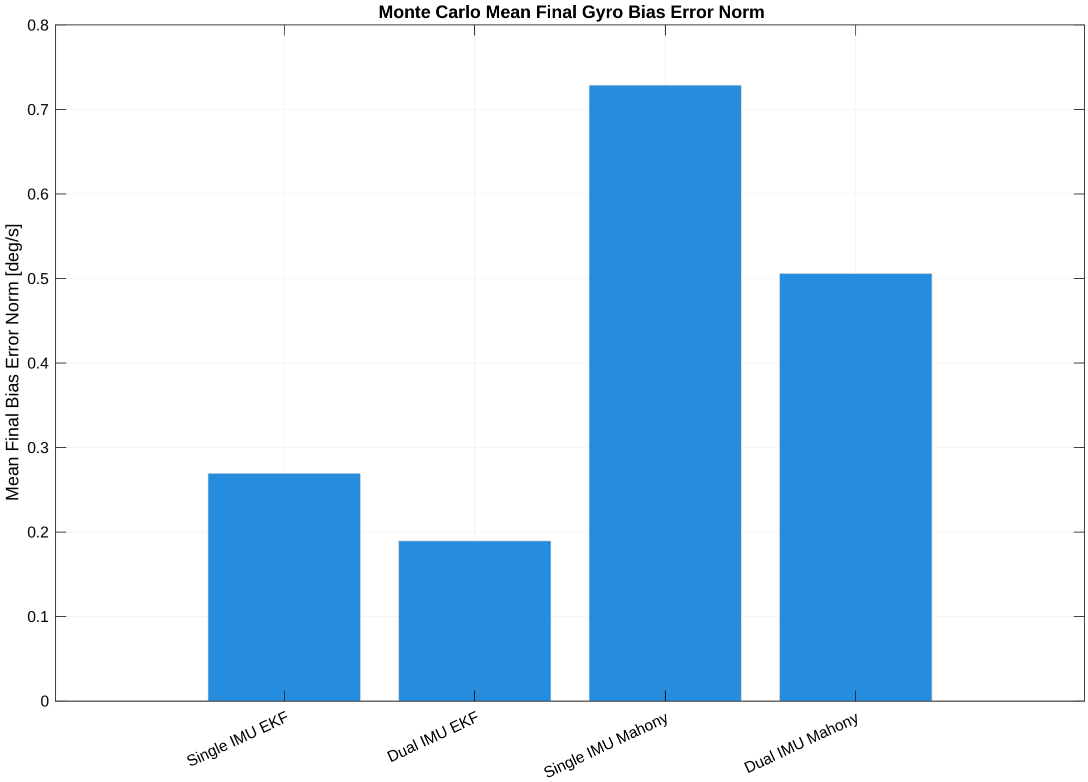


Figure 6: Monte Carlo mean final gyro-bias error norm.

The magnetometer direction error is improved by both calibration and dual fusion. The single-IMU calibrated magnetic direction RMSE is about 0.81°, while the dual fused calibrated value is about 0.62°.

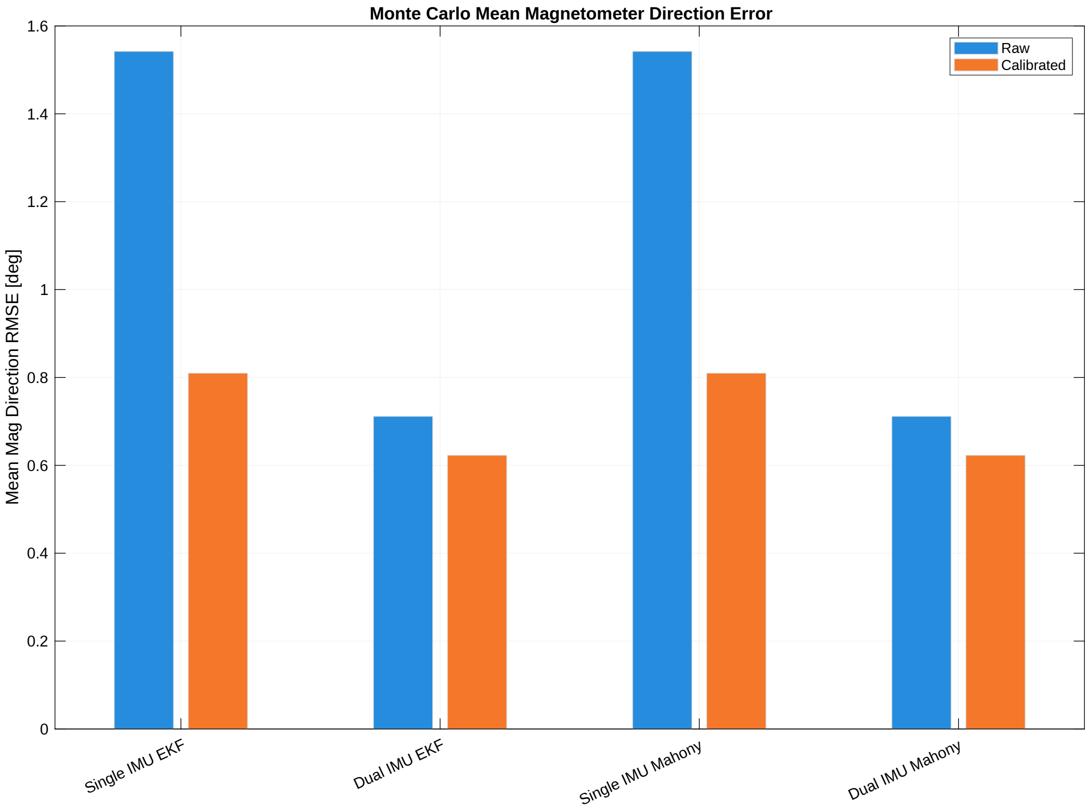


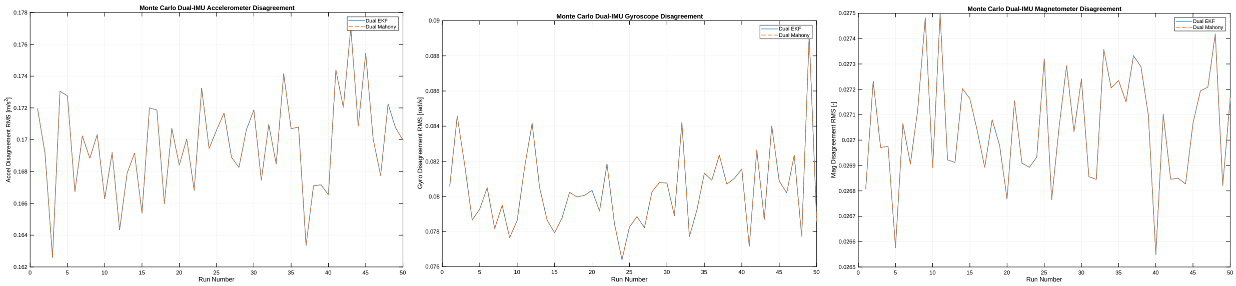
Figure 7: Monte Carlo mean magnetometer direction error before and after calibration.

7.5 Dual-IMU disagreement checks

The dual-IMU disagreement metrics are well behaved. Across Monte Carlo runs, the mean disagreement RMS values are approximately

$$0.1697\ \mathrm{m/s^2}, \qquad 0.0803\ \mathrm{rad/s}, \qquad 0.0271,$$

for accelerometer, gyroscope, and calibrated magnetometer disagreement, respectively. These are nonzero, which means the two IMUs are not duplicates. They are also stable across runs, which means the measurement fusion is not erratic.



(a) Accelerometer. (b) Gyroscope. (c) Magnetometer.

Figure 8: Dual-IMU disagreement metrics over Monte Carlo runs.

7.6 Quaternion normalization

The true and estimated quaternion norm errors are on the order of 2.2×10^{-16}, which is essentially machine precision. This confirms that the quaternion normalization step is working.

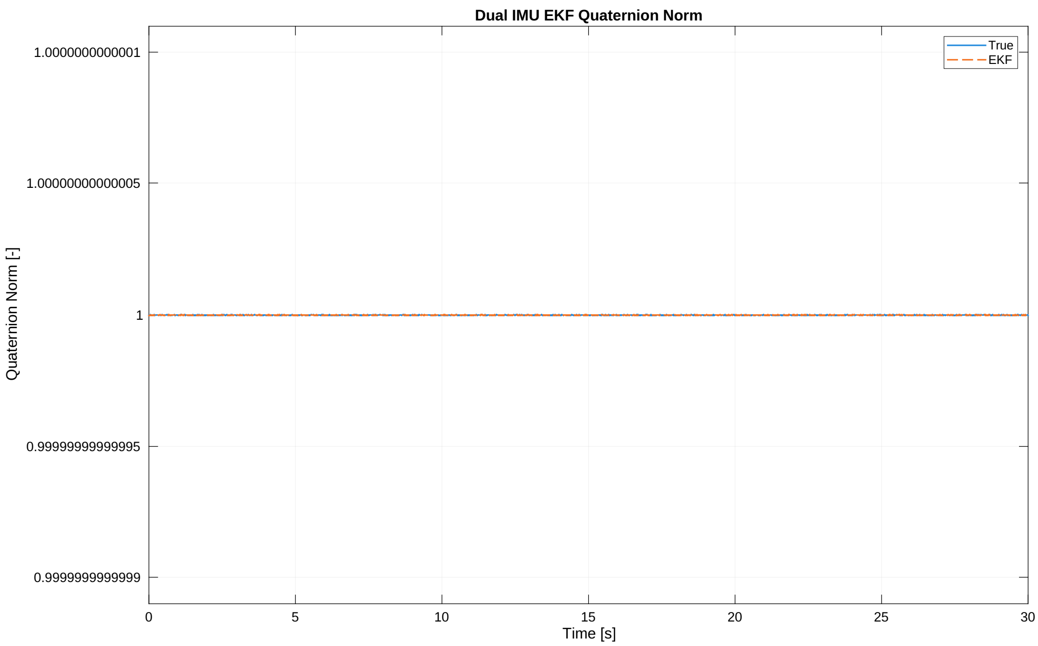


Figure 9: Quaternion norm check for the Dual-IMU EKF case.

7.7 Result verdict

The results are good. The project demonstrates three useful points:

1. The AHRS architecture is valid: gyro propagation is stabilized by accelerometer and magnetometer vector corrections.

2. The dual-IMU measurement fusion improves attitude accuracy for both EKF and Mahony filters.
3. The EKF outperforms Mahony for this noise model and current Mahony gain tuning.

No corrective code changes are required based on the supplied result package. If this were to be extended further, the most valuable next steps would be analytic EKF Jacobians, adaptive measurement covariance, and lever-arm compensation for spatially separated IMUs.

8 References

References

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